DEEPESH SURESH

MECHATRONICS ENGINEERING | ROBOTICS | EMBEDDED SYSTEMS | INDUSTRIAL AUTOMATION

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PROFESSIONAL PROFILE

M.Sc. Mechanical Engineering student at Karlsruhe Institute of Technology (KIT), specialising in Mechatronics, and a B.Sc. Engineering Sciences graduate focused on robotics. Developed underwater ROVs, UAVs, robotic manipulators and ESP32 automation prototypes. Skilled in CAD, embedded systems, sensors, Python/C++ and prototype testing. Internships at Lynxeo and Emerson strengthened engineering validation and problem-solving. Seeking mechatronics, robotics and automation roles.

EDUCATION

Karlsruhe Institute of Technology (KIT) | **M.Sc. Mechanical Engineering** | Current

Karlsruhe, Germany | Specialisation: Mechatronics

École Centrale de Nantes | **B.Sc. Engineering Sciences (Robotics Orientation)** | Sep 2023 – Sep 2026

Nantes, France | Cumulative average: 18.19/20 | 180 validated ECTS

Coursework: Automatic Control, Dynamics, Electrical Engineering, Signal Processing, Numerical Analysis, Programming and FEM.

IIT Patna | **Certificate in Artificial Intelligence & Machine Learning** | Feb – Jul 2026

Online | CGPA: 8.68/10 | Certificate of Excellence in Evaluation

British School of Paris | **A Levels** | Mathematics (A), Physics (A)

ENGINEERING EXPERIENCE

Lynxeo | **Technical Project Intern, Energy Transition & Industrial Cable Systems** | Jun – Aug 2026

Puteaux, France

- Consolidated industrial cable test data, datasheets and technical references to improve engineering traceability.
- Reviewed requirements and deliverables with technical specialists using structured verification checks.
- Proposed a simplified finite-element calculation approach and presented findings to the factory team.

Emerson | **Mechanical Engineering Intern, Industrial Automation** | Jun – Jul 2025

Chennai, India

- Supported industrial automation workflows through technical reviews, documentation and systematic validation.
- Maintained revision-control records, tracked engineering changes and documented technical handovers.

SELECTED MECHATRONICS PROJECTS

Custom 6-DOF Vector-Thrust Underwater ROV | ESP32-S3, Raspberry Pi 4B, Embedded Control

- Co-developed a four-thruster underwater ROV using rotating vector thrusters (0°–90°) for six-axis manoeuvrability; integrated mechanical design, electronics and waterproofing.
- Integrated bidirectional ESC control, Li-ion power, depth hold and stabilisation modes, remote control, Ethernet video streaming and LED lighting.

Pixhawk-Based 4-DOF Underwater ROV | Pixhawk 2.4.8, ArduSub, Sensors

- Designed and built a modular four-thruster ROV based on the Simple ROV-4 architecture, with near-neutral buoyancy for stability and efficiency.
- Configured ArduSub, added water-leak detection and reported approximately 45–60 minutes of endurance depending on operating load.

Hand-Gesture Controlled Pick-and-Place Robot | Arduino Nano, C++, Python

- Built a glove-controlled robotic arm using a flex sensor and SG90 servos; implemented sensor filtering, gesture-to-joint calibration and PWM actuation.
- Integrated hardware, wiring and Python sequencing; demonstrated pick-and-place motions with reported response latency of 120–150 ms.

Smart IoT Water Monitoring & Control System | ESP32, Altium, Java, Android

- Developed an ESP32 controller and Wi-Fi Android app for real-time monitoring of water level, flow rate, volume and motor operation.
- Designed a PCB layout in Altium; assembled a push-button control board, sensor interfaces and 16×2 LCD; tested the water-flow setup.

Custom F450 Quadcopter | APM 2.8, Mission Planner, Brushless Motors

- Assembled an F450 quadcopter integrating an APM 2.8 controller, 1000KV brushless motors, ESC wiring and FlySky radio control.
- Configured and tuned flight parameters in Mission Planner; planned GPS waypoint missions, FPV, LoRa telemetry and custom firmware.

5-DOF Robotic Arm - Mechanical Design | SolidWorks, MG995/MG90S

- Designed a modular five-DOF robotic arm using MG995 servos for primary joints and MG90S micro servos for the wrist and gripper.
- Modelled links, joints, bases and end-effector geometry; constrained and motion-tested the CAD assembly (design/simulation stage).

AI Train Obstacle Detection & Safety Prototype | Computer Vision, AI

- Developed a capstone vision application using track ROI segmentation and multi-class detection of animals, vehicles and debris.
- Combined obstacle location and estimated proximity into Track Clear, Warning and Danger alerts as a research assistance prototype.

Computer-Vision Gesture Interfaces | Python, OpenCV, MediaPipe

- Created real-time hand-tracking interfaces that map camera-detected motion to touchless game-control commands.
- Extended gesture interaction to full-body pose recognition for Subway Surfers and a browser-based spaceship navigation game.

Mechanical Kinematics & CAD Mechanisms | Autodesk Fusion 360

- Modelled and animated Geneva, Scotch Yoke, Trammel of Archimedes and wheel mechanisms to study indexing and constrained motion.
- Applied assembly joints and motion limits to demonstrate reciprocating, elliptical and rotary mechanism behaviour.

Smartphone Motion-Controlled Interactive Game | Python, IMU, WebSockets

- Developed a fruit-slicing game controlled by real-time smartphone gyroscope and accelerometer data.
- Streamed IMU data over a local network and mapped sensor angles to on-screen coordinates for touchless gameplay.

NeuroPlay Algorithmic Game Arcade | JavaScript, AI Search, Game Theory

- Built a modular multi-game arcade with vanilla HTML, CSS and JavaScript, reusable interfaces and deterministic state management.
- Implemented Minimax with Alpha-Beta Pruning for Connect 4, misère strategy for Pop It and risk-aware Dots & Boxes AI.

TECHNICAL SKILLS

Robotics & embedded: ESP32/ESP32-S3, Arduino Nano, Raspberry Pi, Pixhawk/APM, ArduSub, Mission Planner, ROS, PWM/ESC, sensors, controller tuning

CAD & simulation: SolidWorks, Fusion 360, CATIA V5, AutoCAD, FEM, MATLAB, mechanism design, assembly constraints, validation

Programming & HCI: Python, C++, Java, OpenCV, MediaPipe, SQL, JavaScript, Git, Linux, Android Studio, real-time sensor processing

Languages: Tamil (native), English (fluent), French (B1), German (learning)